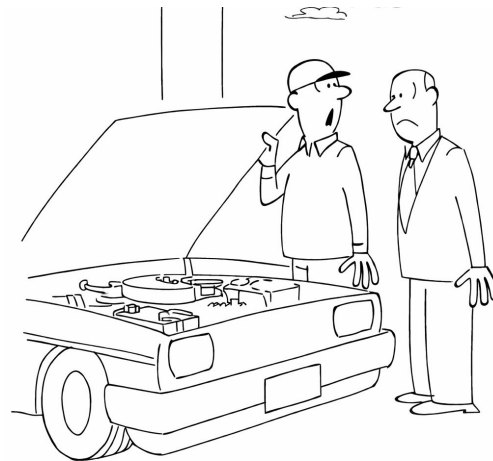


AutoRisk

Predicting High-Risk Vehicles from NHTSA



Ning Mu
STATS 418 • Spring 2026 • Final Project Proposal

Some car buyers don't know what's wrong before they buy.



50K+

*consumer complaints filed to NHTSA every
year*

free, public, and almost nobody uses them

Problem

Reliability data exists in NHTSA recall + complaint databases, but is fragmented across 50+ endpoints and millions of records.

Goal

Predict whether a (year, make, model) falls in the top 30% of complaint rate, using only metadata + historical recall signals.

Why it matters

Helps buyers, insurers, and dealers screen vehicles. Demonstrates an end-to-end ML pipeline (API → model → API → app) with real, messy public data.

1. Data Sources

NHTSA public APIs :

- vehicle catalog
- Recalls
- complaints

2. Engineering Design

- Local caching to avoid repeated API calls
- Rate limiting and retry mechanism

3. Data Issues & Fixes

- Duplicate trim-level entries detected
- Merged into base models to avoid double-counting

Raw Data

F-150 SuperCab
F-150 Crew Cab
Duplicate complaints

Normalize

→ base_model
SuperCab → F-150
Crew Cab → F-150

Group

year + make + base_model

Aggregate

Complaint → MAX
Recall → SUM

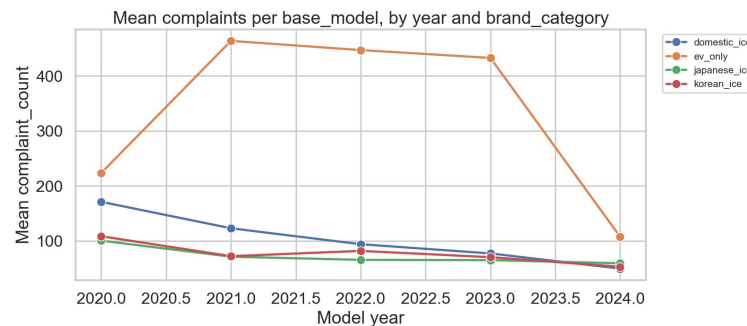
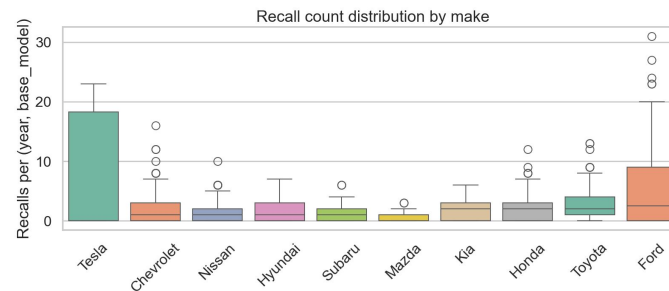
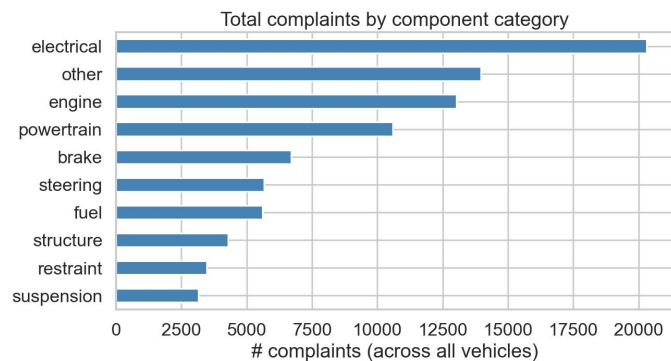
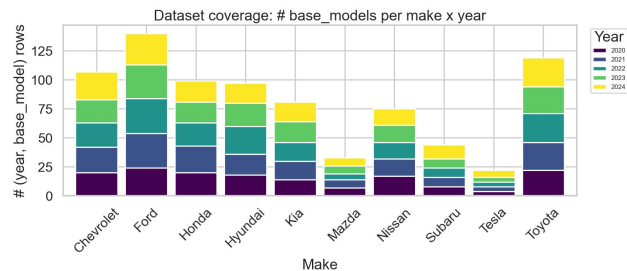
Final Dataset

1 row per vehicle
clean features
complaint_rate

Current Progress

Merged dataset: 10 brands, recent 5 years (2020-2024)

03 — EXPLORATORY DATA ANALYSIS



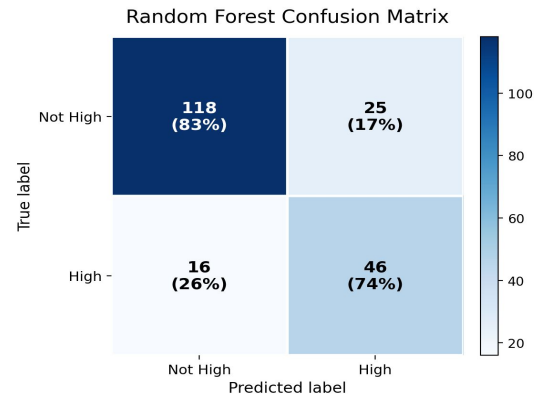
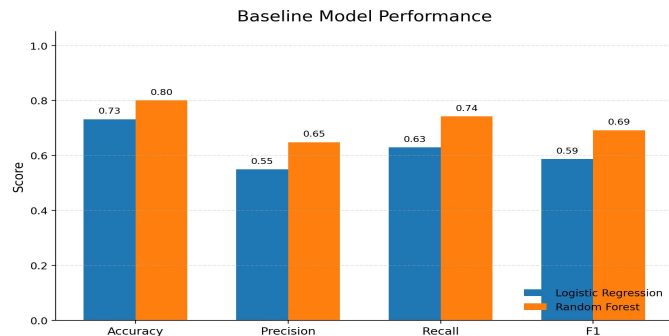
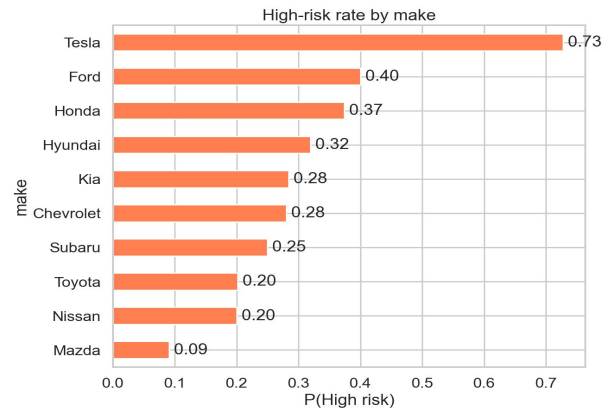
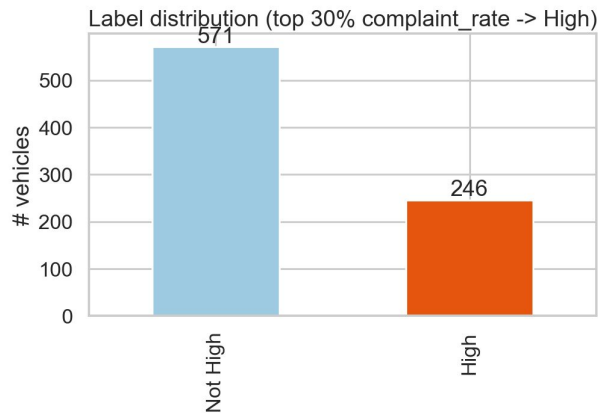
Insight 1. Data imbalance

Insight 2. Recall variability

Insight 3. Key issue: electrical systems

Insight 4. EV risk trend

04 — MODELING APPROACH

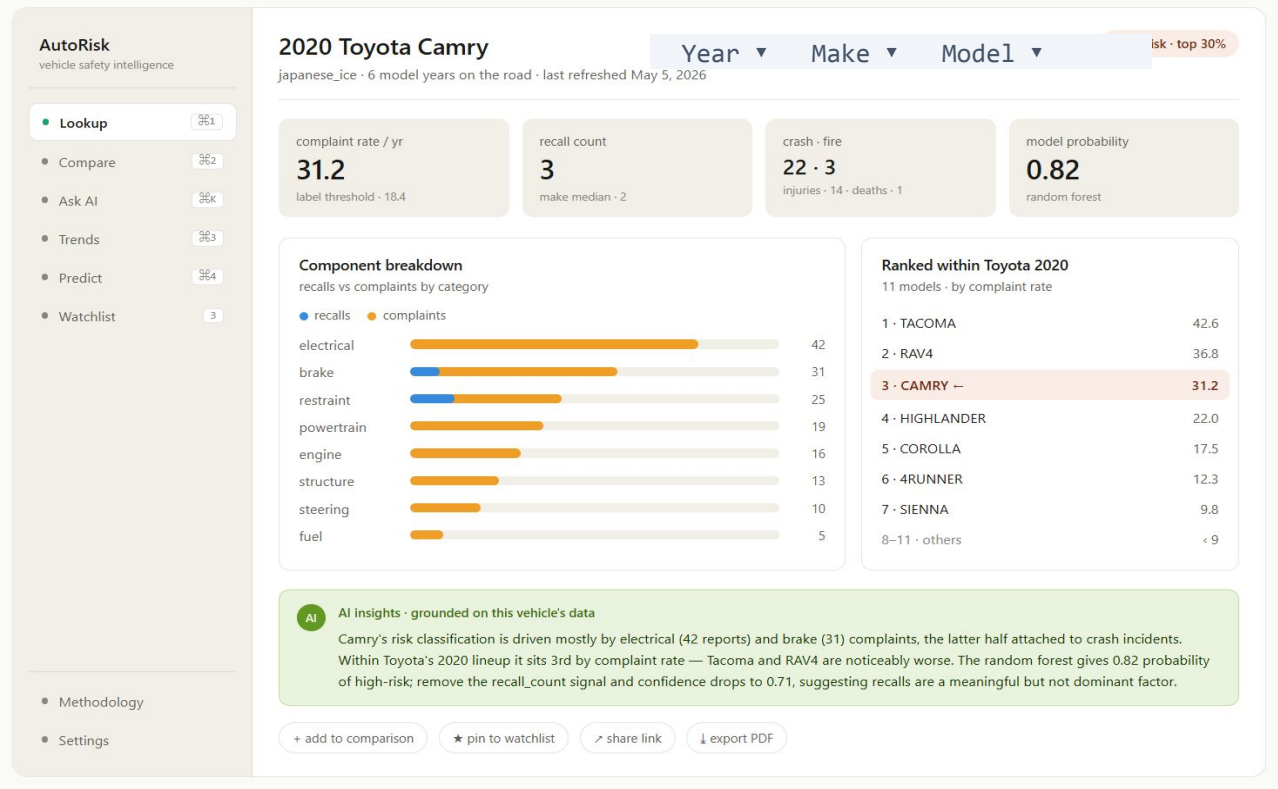


Insight 1. The dataset has a moderate class imbalance

Insight 2. Random Forest is better

Insight 3. Risk varies significantly across different brands

Insight 4. The model achieves good performance in identifying both high-risk and non-high-risk vehicles.

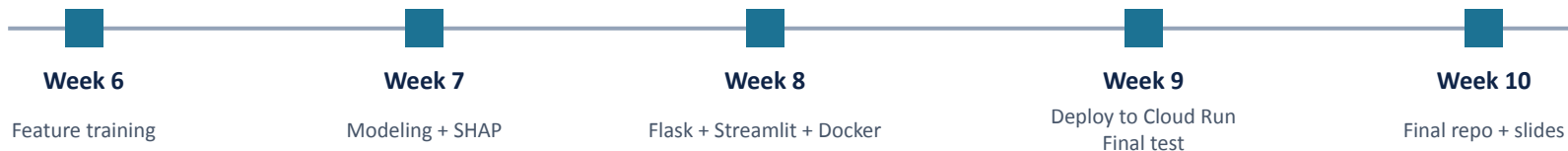


Features used for modeling:

- Vehicle metadata
 - year, vehicle_age
 - make, brand_category
- Recall-based features
 - total recall count
 - recall counts by component category
- Aggregated features
 - trim-level merged base model

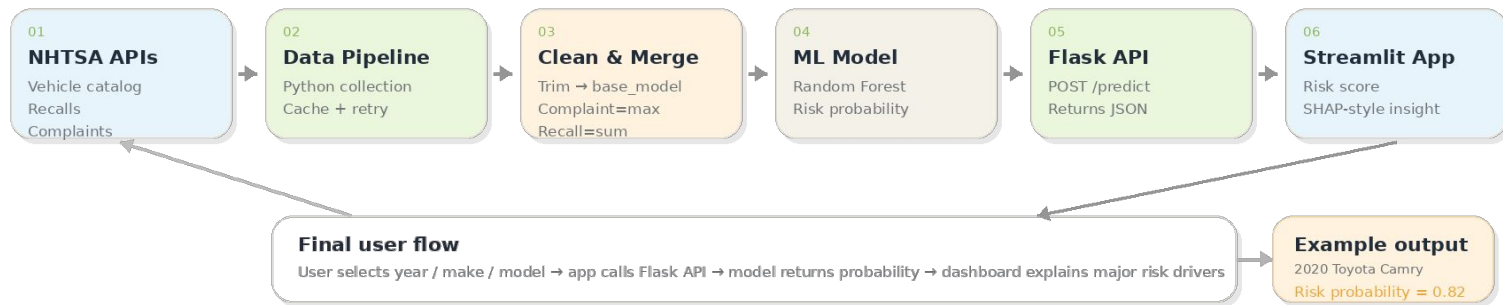
Complaint-based features are excluded to avoid data leakage.

TIMELINE



AutoRisk System Architecture

Data → features → prediction API → web dashboard



Potential Challenge :

1. Too many feature.
3. render cold start

2. Ai model training